How DLinear makes a forecast

Zeng et al. (2023) · supervised baseline trained on each dataset · Darts DLinearModel

1. A window of inputs

Each forecast reads a window of the last 400 days (36 months). A ? is a value not known yet.

past *future*

Target temperature

Known ahead day length

Past only NAO index

Prices go in as log differences, oil and yields as differences; all inputs are min-max scaled on the training rows.

2. Split trend and remainder

A moving average over the window gives each row its trend; what is left is the remainder.

window

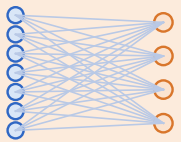
trend (moving average)

remainder

Average length: tuned among 7, 25 and 61 days (3, 7 and 13 months).

3. Two linear layers

One linear layer maps the whole trend window to the horizon, another the remainder; the two outputs are added.



window *horizon*

$$\hat{y} = W_T T + W_R R + W_x x$$

T and R are the trend and remainder of every input row; x is the known-ahead covariates at each future step.

4. Train once, reuse

Trained once per dataset on the rows before the first test window (daily: the last 4,096), then every window is forecast from its own origin with no retraining.

point forecast

Optuna tries a few settings on validation windows inside the training rows; the first is the default. 15 epochs, MSE loss.

A strong, simple yardstick: if a foundation model cannot beat two linear layers, it adds little.

Window 400 days or 36 months · direct forecast of 90 days or 24 months at once · tuned: average length, learning rate